CNC Lathe CL-200 - Maintenance Manual (sample)

1. Daily checks

Before the first shift, check the spindle oil level in the sight glass on the headstock; it must be between the two marks. Use only ISO VG 32 spindle oil. Check that the hydraulic chuck pressure gauge reads between 22 and 28 bar. Check the coolant level in the tank; the tank holds 120 litres. Wipe the way covers and remove chips from the chip conveyor.

2. Lubrication schedule

Slideways are lubricated automatically with ISO VG 68 way oil; refill the way-oil reservoir (2 litres) when the low-level lamp is on. Change the spindle oil every 2000 operating hours. Grease the tailstock quill every 200 operating hours with lithium grease NLGI 2. Change the hydraulic oil (ISO VG 46, 40 litres) every 4000 operating hours and replace the hydraulic return filter at the same time.

3. Alarm codes

Alarm E-204: spindle overheating. The spindle temperature has exceeded 70 degrees C. Stop the spindle, check that the coolant pump CP-12 is running, clean the coolant filter and check the spindle oil level. Do not restart until the temperature is below 45 degrees C.
Alarm E-310: chuck pressure low. The chuck pressure is below 18 bar. Check the hydraulic oil level and the pressure relief valve setting.
Alarm E-415: turret index fault. The turret did not lock within 3 seconds. Check the turret proximity sensor and clean chips from the turret base.

4. Coolant system

The coolant concentration must be kept between 6 and 8 percent, measured with a refractometer once a week. Clean the coolant filter every 50 operating hours. Replace the coolant completely every 6 months. The coolant pump CP-12 delivers 30 litres per minute; a delivery below 20 litres per minute means the pump or the filter needs attention. Spare parts: coolant pump CP-12, coolant filter element CF-08, spindle belt SB-1450.